

Appendix A – Suggested Tkinter Widgets

The following widgets are commonly used when developing graphical user interfaces in Python. This appendix is provided only as guidance. You are free to use additional widgets if required.

Widget	Typical Use
Tk	Creates the main application window.
Frame	Groups related widgets into logical sections.
Label	Displays static text such as titles, usernames, and status messages.
Entry	Accepts single-line user input such as usernames or chat messages.
Button	Performs actions such as Connect, Send, Disconnect, or Exit.
Listbox	Displays the list of online users.
ScrolledText	Displays chat messages with a vertical scrollbar.
Scrollbar	Provides scrolling support for text areas or lists.
Messagebox	Displays information, warning, confirmation, or error dialogs.
Menu (Optional)	Creates application menus such as File, Help, or Exit.
StringVar / BooleanVar	Stores values associated with GUI widgets.

General Suggestions

- A simple and functional interface is sufficient.
- Focus on correct functionality rather than appearance.
- Organize widgets neatly using Frames.
- Keep the interface responsive while messages are being received.
- Test the application using multiple clients in Mininet.